SC

(10 & A-11) SEAT No.________ **No. of Printed Pages : 3**

SARDAR PATEL UNIVERSITY

BCA SEM-I EXAMINATION (Reg.&NC)

2017

TUESDAY, 7TH NOVEMBER

10:00 am to 12:00 noon

US01FBCA02: MATHEMATICS-I **Total Marks: 70**

Q:1 Choose the correct option in the following, mention the correct option with the answers in the answer book. [10]

(1) Median of 4, 7, 3, 11, 17, 14, 9, 8

(a) 9.5 (b) 9 (c) 8.5 (d) 8

(2) A Square matrix A is said to be symmetric if.....

(a) $A \neq A^T$ (b) $A = -A^T$ (c) $A = A^T$ (d) None

(3) The number of elements in the power set of a set { a, b, c, d } are:

(a) 4 (b) 16 (c) 8 (d) 32

(4) Let A = {1, 0}, then A closed under:

(a) multiplication (b) addition (c) Division (d) Subtraction

(5) The identity for a group $(R-\{0\}, \times)$ is:

(a) 1 (b) 0 (c) −1 (d) e

(6) Geometric mean of 2 , 3 and 6 is

(a) 6 (b) 3.67 (c) 3 (d) 2

(7) The set $\{x \in Q: 2 < x < 7\}$ is:

(a) finite (b) Infinite (c) Empty (d) none

(8) If $f(x) = 2x - 3$, then $f^2(1) =$

(a) −1 (b) 5 (c) −5 (d) $4x - 9$

(9) Every monoid are:

(a) group (b) ring (c) semigroup (d) none

(10) 4(1, 2, 1) + 2(1, 3, 3) = ...

(a) (6 , 14 , 10) (b) (6 , 10 , 14) (c) (6 , 14 , 10) (d) (10 , 6 , 14)

Q:2 Answer the following in short (**Attempt any Ten**). [20]

(1) Define: Ring and Unity of a ring.

(PTO)

(2) Find the inverse of the matrix $\begin{bmatrix} 3 & 7 \\ 2 & 5 \end{bmatrix}$.

(3) Find Median height(in cm) of seven students for the following data
150, 165, 154, 156, 159, 145, 157

(4) Obtain mean of observations 3, 5, 6, 10, 4, 7, 9, 12 and 10.

(5) If S is a nonempty set with the operation x * y = x. Is the operation:
(i) associative?, (ii) commutative ?.

(6) Find the power set of a set {1, 2, 3}.

(7) If $A = \begin{bmatrix} 2 & 0 & -1 \\ 4 & 5 & 3 \\ 0 & 2 & 5 \end{bmatrix}$ then find $A + A^T$ and $A - A^T$.

(8) Define qualitative data.

(9) Find x, y, z if (2x, 3, y) = (4, x + z, 2z).

(10) In $(Z_7, \times_7)$, find 3^{-1}, 6^{-1} ,if exists.

(11) If f(x) = x + 3 and g(x) = 3x +1 then find fog(0) and gof(−1) .

(12) For a, b rational number, define a*b = ab/2. Find identity element for given binary operation.

Q:3

(a) Let n denote a positive integer. Suppose a function L is defined as [5]

$L(n) = \begin{cases} 0 & if\ n = 1 \\ L(\lfloor n/2 \rfloor) + 1 & if\ n > 1 \end{cases}$. Find L (25) and L (34) .

(b) Define invertible function and hence find inverse of the function [5]
$f(x) = \dfrac{7x-3}{5x-2}$, $x \neq \dfrac{2}{5}$.

Q:3 **OR**

(c) Prove that $1^2 + 2^2 + 3^2 + \dots\dots + n^2 = \dfrac{n(n+1)(2n+1)}{6}$. [5]

(d) By using algebra of sets , prove that $(\phi \cup A) \cap (B \cup A) = A$. [5]

Q:4

(a) If G = {1,2,3,4,5,6} then prove that G is a group under multiplication modulo 7. Is it finite group? [5]

(b) Define a group homomorphism. Show that f: G →G′ defined by $f(a) = 2^a$ is a homomorphism where G is a group of real numbers under addition and G′ is a group of positive real numbers under multiplication. [5]

Q:4 **OR**

(c) For a,b ∈ Q (rational numbers), define a*b = ab/3. [5]

(i) Is (Q,*) Semigroup? (ii) Is (Q,*) Monoid ?

(iii) Find the inverses of elements of (Q,*), if it exist.

(d) For a,b rational number, define a*b = a + b - ab. Is (Q,*) commutative? Show that (Q,*) is Monoid and find its inverse if it exist. [5]

Q:5

(a) Using Cremer's rule solve the following simultaneous equations [5]

3x - 2y = 5, 5x +4y =1.

(b) If $A = \begin{bmatrix} 1 & -2 & 3 \\ 6 & 0 & 9 \\ 5 & -7 & 11 \end{bmatrix}$ then find $A + A^T$, $A - A^T$ and $A A^T$ [5]

Q:5 **OR**

(c) Let $\vec{u} = (-3, 4, 8)$ and $\vec{v} = (2, -5, 4)$. Find: [5]

(i) $\vec{u} - 2\vec{v}$ (ii) $3\vec{u} - 5\vec{v}$ (iii) $\|\vec{u}\|$ (iv) $\|\vec{v}\|$ (v) $\|4\vec{u} - \vec{v}\|$

(d) If $A = \begin{bmatrix} 2 & 4 \\ 3 & 0 \\ 3 & 2 \end{bmatrix}$, $B = \begin{bmatrix} 4 & 2 & 3 \\ 7 & 1 & 5 \end{bmatrix}$ then prove that (AB)' = B'A'. [5]

Q:6 Calculate Mean, Median and Mode for the following data. [10]

weight(lbs) X	130	135	140	145	146	148	149	150	157
no. of persons(f)	3	4	6	6	3	5	2	1	1

OR

Q:6 The marks of 40 students who attended a workshop competitive exam are as follows: [10]

27	32	57	34	36	48	49	31	51	34
49	45	51	29	47	36	50	46	30	46
35	35	48	41	53	36	37	47	47	30
43	45	42	30	46	50	28	44	48	49

Classify the above data in exclusive classes & one of them being 40 - 44. Also obtain mean of the distribution.

—x—